SAC Soft Actor Critic

TD3 Twin Delayed Deep Deterministic Policy Gradient

MoE Mixture of Experts

DDQN Double Deep Q-Network

TD Temporal Difference

RL-Course 2024/25: Final Project Report

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1 Introduction

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2 Method

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2.1 Soft Actor Critic

Our Soft Actor Critic (SAC) implementation is based on the original paper by [3] and its follow-up paper [4]. SAC is an off-policy actor-critic algorithm that learns a stochastic policy in an environment with continuous action and state spaces. The algorithm is based on the principle of maximum entropy reinforcement learning, which aims to maximize the expected reward while also maximizing the entropy of the policy. This encourages exploration and prevents the policy from getting stuck in local optima. The major components of the SAC algorithm are the actor, the critic, and the entropy target:

$$J(\pi) = \sum_{t=0}^{T} \mathbb{E}_{(s_t,a_t)\sim\rho_\pi} \left[r(s_t, a_t) + \alpha \mathcal{H}(\pi(\cdot \mid s_t)) \right] \tag{1}$$

The actor is a stochastic policy network, which is based on a vanilla neural network with a Gaussian distribution output layer. It estimates the mean and standard deviation $\mu(s), \sigma(s) = f_\theta(s)$ and provides a sampling method which returns both a sample from the distribution and the log probability of the selected action. To make the sampling process differentiable, the reparameterization trick is used where $z \sim \mathcal{N}(0, 1)$ and $u = \mu(s)+\sigma(s)\cdot z$. The action bounds are enforced by applying the $\tanh$ function to the sampled unbounded action. To consider this bounding in the probability density function, the probability density function of the unbounded action is scaled by the absolute determinant of the Jacobian matrix of the $\tanh$ function [1][C. Enforcing Action Bounds]:

$$\pi(a \mid s) = \mu(u \mid s) \left| \det \left(\frac{d\mathbf{a}}{d\mathbf{u}} \right) \right|^{-1}. \tag{2}$$

So for each state, sampled from the replay buffer $\mathcal{D}$ an action is sampled from the actor and the log probability of this action is calculated using the adjusted probability density function. The actor is trained using the maximum entropy objective, where the critic estimates the value for each state-action pair:

$$J(\pi) = \mathbb{E}_{s_t\sim\mathcal{D}, a_t\sim\pi} \left[\alpha \log \pi(a_t \mid s_t) - Q(s_t, a_t) \right] \tag{3}$$

The critic consists of two Q-networks, valuing state-action pairs, to prevent overestimation bias [2]. During training both networks independently predict a value for the given state-action pair $(s_t \sim \mathcal{D}, a_t \sim \pi)$ and the minimum is selected.
The target Q-values incorporate the immediate reward r and the discounted future reward. Additionally, an entropy term, weighted by the temperature parameter α, is used to balance exploration and exploitation. This formulation aligns with the objective function $J(\pi)$, which maximizes both the expected reward and the entropy of the policy:

$$y = r + \gamma(1-d)\left(\min\left(Q_1^{\text{target}}(s', a'), Q_2^{\text{target}}(s', a')\right) + \alpha\mathcal{H}(\pi(\cdot \mid s'))\right). \tag{4}$$

The critic is trained by minimizing the mean squared Bellman error, which represents the difference between the estimated Q-values and the target values:

$$\mathcal{L}_Q = \mathbb{E}\left[(Q(s,a) - y)^2\right]. \tag{5}$$

To stabilize the training process, the target Q-networks are updated using a soft target update mechanism, as it was shown to improve the stability of the training process [3][E. Additional Baseline Results].
In the original paper "an additional function approximator for the value function [was introduced] but later found [as] to be unnecessary" [4]

Moreover the automatically temperature parameter tuning using gradient-based optimization, introduced in the follow-up paper [4], was implemented:

$$J(\alpha) = \mathbb{E}a_t \sim \pi\left[-\alpha\left(\log \pi(a_t \mid s_t) + \mathcal{H}\text{target}\right)\right], \tag{6}$$

where $\mathcal{H}$target is set to negative of the action space dimensionality, as proposed in [4][D. Hyperparameters]

2.2 Hybrid-Model

While reviewing the gameplay of the trained SAC and Twin Delayed Deep Deterministic Policy Gradient (TD3) agents, we noticed that both agents have their strengths and weaknesses in different situations. To combine the strengths of both agents, we created a hybrid model. This approach is inspired by the Mixture of Experts (MoE) architecture, where multiple experts are combined using a gating network to weight the experts' predictions. But since we've got an continious spatial action space, weighting both 'experts' predictions would not be feasible option. Thus a Double Deep Q-Network (DDQN) [5] was implemented as selection gate for the actions. This gate values the predicted actions of both agents and select the best action for each state:

$$a_{\text{Hybrid}} = \arg \max_{a \in \{\pi_{\text{SAC}}(s), \pi_{\text{TD3}}(s)\}} Q_{\text{DDQN}}(s, a) \tag{7}$$

The loss function for training the DDQN is based on the Temporal Difference (TD)-Error, which is the difference between the predicted and the target Q-value. As a major difference to the MoE approach the 'expert' models are fixed (trained independently) and not trained simulataneously to the gating network.

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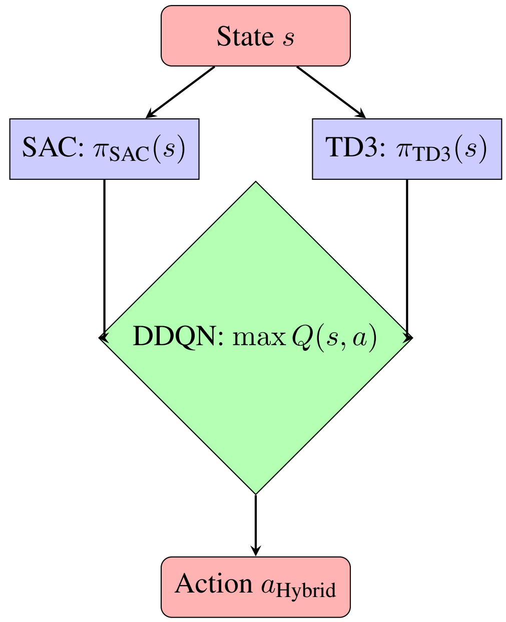

Figure 1: Hybrid-Model with DDQN Selection Gate

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